Computer Vision

Assignment 4

This is the fourth assignment for your computer vision course. After finishing this assignment, you are expected to know:

- Color Image Processing
- Image enhancement using Morphological operators

Instructions

- Please provide a proper answer for each part of the questions.
- Use the cells beneath each question for your answer.
- Feel free to add more cells whenever needed.

Good luck!

Question 1 (15 points)

In this problem, you will use your image processing knowledge to repair a corrupted image. Remove the lines and artifacts from the corrupted *“fruits.png”* image.



fruits.png

Question 2 (15 points)

Apply histogram equalization to tune the low-exposure image *“badexposure.jpg”*.

Show the histograms before and after applying the histogram equalization algorithm.

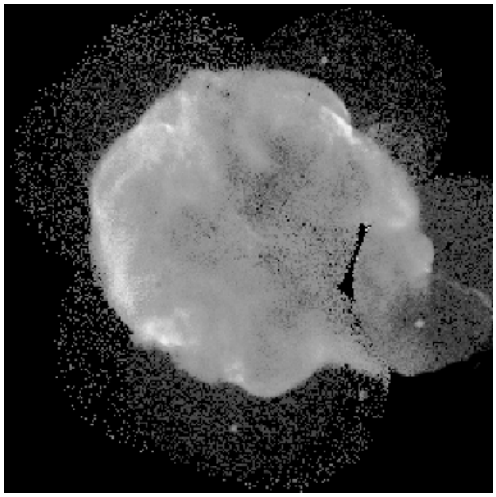


badexposure.jpg

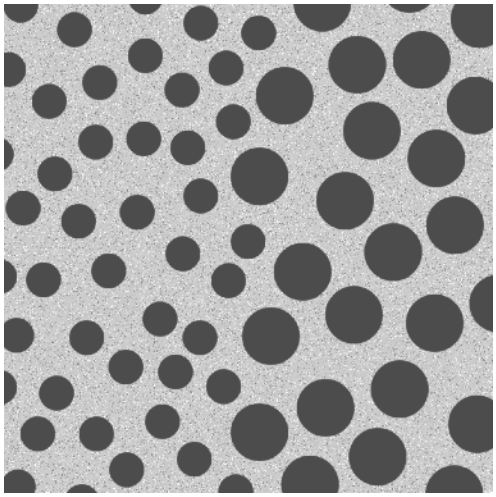
Question 3 (20 points)

A) Remove the noise from the image *“splash.png”*.

B) Separate the two types of blobs in the *“blobs.png”*.



splash.png



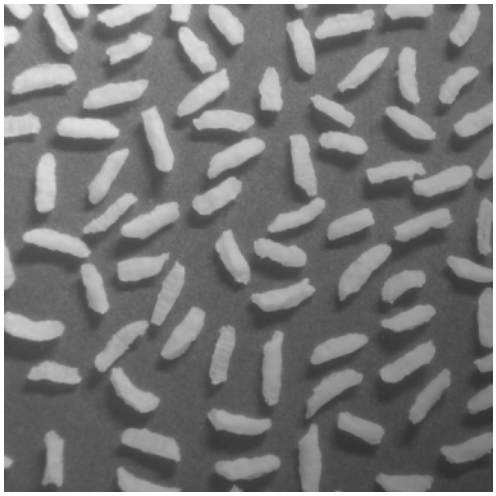
blobs.png

Question 4 (20 points)

The image *“rices.png”* includes objects of rice grains of different orientations over a dark background. In this problem, you are asked to develop a program to:

A) Detect (extract from the background) the rice grain objects (By example, using thresholding)

B) Compute the number and average size of the rice grains. (Use the morphological operations to count the number of rice grains, such as Opening, Closing connected components, and …)



rices.png